Track: Multi-class classification (4 classes)
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Data Description

- **Source:** Intel Image Classification dataset (Kaggle): https://www.kaggle.com/datasets/puneet6060/intel-image-classification
- **Classes (4):** forest, glacier, sea, street
- **Training data** (from seg_train):
  - Dataset A: 60 images per class = 240 images total
  - Dataset B: 100 images per class = 400 images total
- **Hold-out split:** 80% training, 20% testing, shuffled randomly each run. Dataset A gives 192 train / 48 test. Dataset B gives 320 train / 80 test.
- **Manual test set:** 40 unseen images from seg_test (10 per class), classified one by one in the app.

Preprocessing

1. Resize each image to 16×16 or 32×32.
2. Scale the R, G and B values from 0–255 to 0–1.
3. Flatten into 768 inputs (16×16×3) or 3,072 inputs (32×32×3).
4. Use a one-hot target (1 for the correct class, 0 for the others).

Method

- **Network:** 3 layers using the provided INeuron, HNeuron and ONeuron classes. The hidden layer has 16, 32 or 64 neurons, the output layer has 4, and all activations are sigmoid.
- **Training:** backpropagation with one weight update per randomly picked training image.
- **Loss:** training MSE, logged every 10 epochs.
- **Prediction:** the output neuron with the highest value wins.
- **Environment:** Windows Forms, .NET Framework 3.5, Visual Studio.

Experiments

Run	Images/class	Size	Inputs	Best config (hidden / lr / epochs)	Internal test accuracy	Manual test (40)
1	60	16×16	768	64 / 0.2 / 200	68.8% (33/48)	21/40 (52.5%)
2	100	16×16	768	32 / 0.1 / 100	53.8% (43/80)	20/40 (50.0%)
3	100	32×32	3,072	16 / 0.1 / 100	52.5% (42/80)	24/40 (60.0%)

In run 2, two configs tied at 53.75%, and the program kept the first one.

Hyperparameter Tuning

Each run tested the same 5 configurations. The values are test accuracy.

Hidden	LR	Epochs	Run 1 (60, 16×16)	Run 2 (100, 16×16)	Run 3 (100, 32×32)
16	0.1	100	58.3%	51.3%	**52.5%**
32	0.1	100	66.7%	**53.8%**	47.5%
32	0.2	100	64.6%	52.5%	51.3%
32	0.2	200	64.6%	51.3%	48.8%
64	0.2	200	**68.8%**	**53.8%**	51.3%

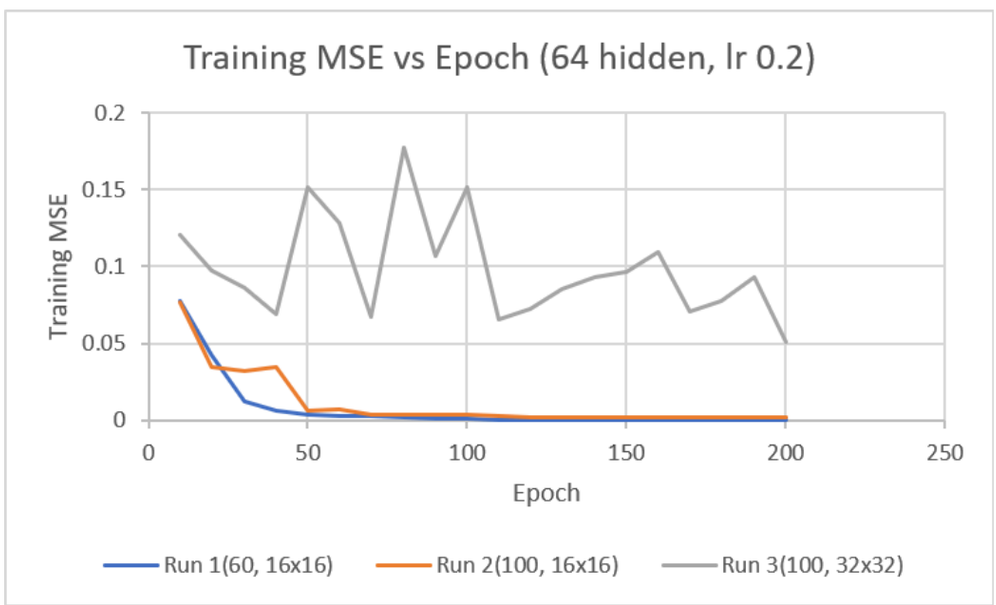
- **Hidden size:** 16 neurons was the worst in run 1, and 64 was the best.
- **Epochs:** 200 epochs was no better than 100, because the loss was already near zero.
- **Learning rate:** 0.1 and 0.2 gave similar results.
- **Runs 2 and 3:** the configs differ by only a few points, which is within noise for 80 test images.

Training Log (loss decreasing)

Training MSE at selected epochs for the best configs (from the log CSVs):

Run	Config	Epoch 10	Epoch 50	Epoch 100	Epoch 200
1	64 / 0.2	0.0773	0.0039	0.0006	0.0002
1	32 / 0.1	0.1194	0.0234	0.0037	–
2	32 / 0.1	0.1108	0.0150	0.0036	–
2	64 / 0.2	0.0768	0.0065	0.0035	0.0017
3	16 / 0.1	0.1041	0.0443	0.0409	–
3	64 / 0.2	0.1204	0.1514	0.1517	0.0509

Loss chart:



Evaluation

Per-class precision and recall (internal test set)

Class	Run 1 P / R	Run 2 P / R	Run 3 P / R
forest	0.75 / 0.60	0.77 / 0.77	0.67 / 0.53
glacier	1.00 / 0.63	0.42 / 0.61	0.46 / 0.65
sea	0.44 / 0.89	0.41 / 0.33	0.33 / 0.29
street	0.70 / 0.70	0.53 / 0.42	0.64 / 0.58

Manual test (40 unseen images, correct out of 10)

Class	Run 1	Run 2	Run 3
forest	8	6	8
glacier	2	5	6
sea	6	3	3
street	5	6	7
Total	**21 (52.5%)**	**20 (50.0%)**	**24 (60.0%)**

Main mistakes in the manual test

- **Run 1:** Glacier images were predicted as sea 6 times, and sea as glacier 4 times.
- **Run 2:** Sea images were predicted as glacier 7 times, and glacier as sea 3 times.
- **Run 3:** Sea images were predicted as glacier 7 times, and forest as street 2 times.

Analysis

- **Overfitting:** the training loss drops to almost zero, but test accuracy is only 50–69%. This is still well above the 25% chance level.
- **Sea and glacier are confused:** both have similar blue and white colors, and the network only sees color values, not shapes or textures.
- **Dark or brown images are predicted as street:** this affected some forest images and glacier images showing mountains with little snow.
- **Image size:** 32×32 did not clearly improve accuracy, and its loss stayed higher. The larger input may slow learning, but we did not test this.
- **Dataset size:** more images per class did not raise accuracy. The runs use different random splits and test sizes, so no firm conclusion can be made.

Limitations

- Each setup was run only once, so results would vary between runs.
- The test sets are small, so a difference of a few points is not meaningful.

Conclusion

The 3-layer network learns to classify the 4 image classes, reaching 52–69% test accuracy compared with 25% by chance. A very small hidden layer (16) underfits, and extra epochs do not help once the loss is near zero. The main limits are overfitting and confusion between similar classes like sea and glacier